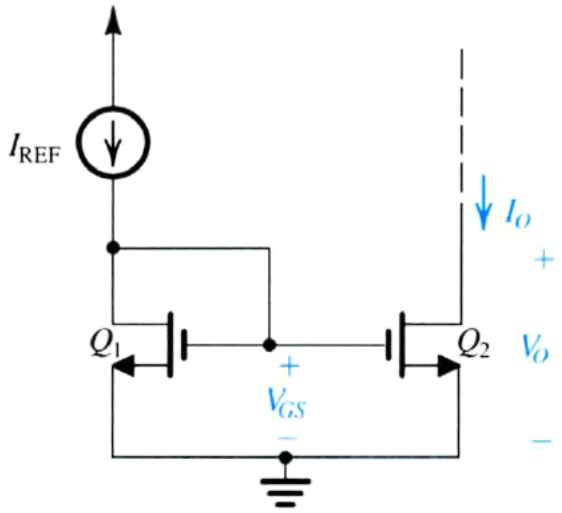
Reading

Adel S. Sedra and Kenneth C. (KC) Smith, Tony Chan Carusone, and Vincent Gaudet
Microelectronic Circuits 8th Edition, *Oxford University Press*, 2019.

- Chapter 8.6 The IC Source Follower
- Chapter 8.7 Current Mirror Circuits with Improved Performance

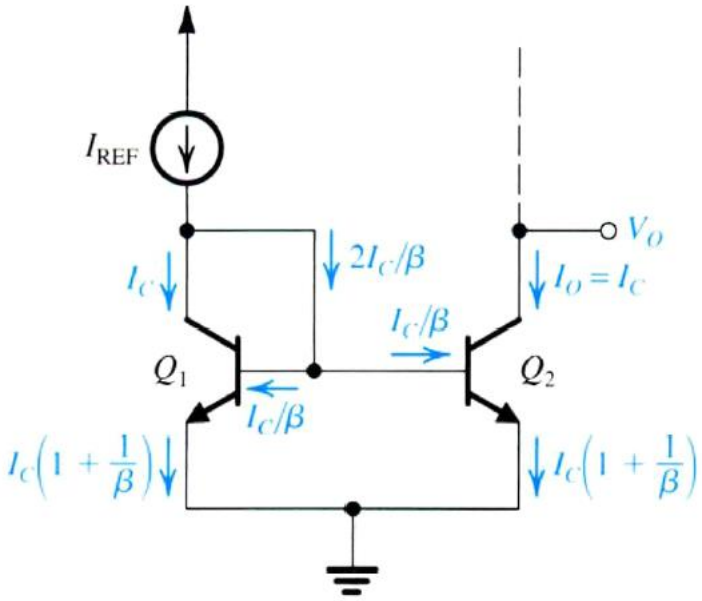
Recap



$$I_O|_{\text{nominal}} = \frac{(W/L)_2}{(W/L)_1} I_{REF}$$

$$I_O = \frac{(W/L)_2}{(W/L)_1} I_{REF}\left[1 + \frac{V_O - V_{GS}}{V_{A2}}\right]$$

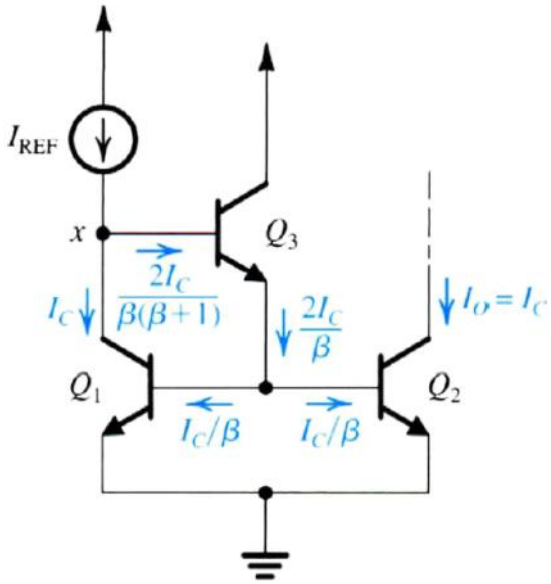
$$R_o = r_{o2}$$



$$\frac{I_O}{I_{REF}} = \frac{I_C}{I_C\left(1+\frac{2}{\beta}\right)} = \frac{1}{1+\frac{2}{\beta}}$$

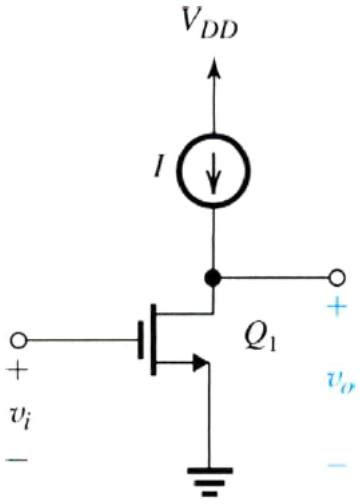
$$\frac{I_O}{I_{REF}} = \frac{m}{1+\frac{m+1}{\beta}}$$

$$I_O = I_{REF}\frac{m}{1+\frac{m+1}{\beta}}\left(1+\frac{V_O - V_{BE}}{V_{A2}}\right)$$

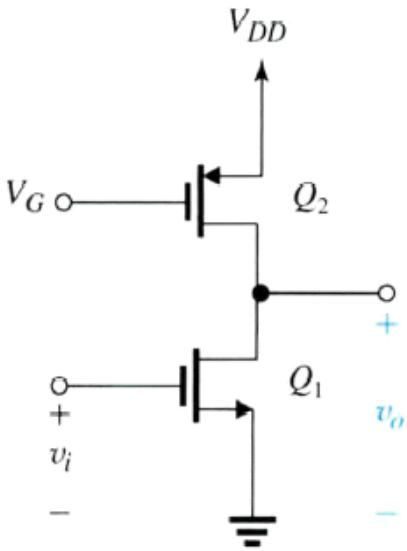


$$\frac{I_O}{I_{REF}} = \frac{1}{1+2/(\beta^2+\beta)} \simeq \frac{1}{1+2/\beta^2}$$

Recap



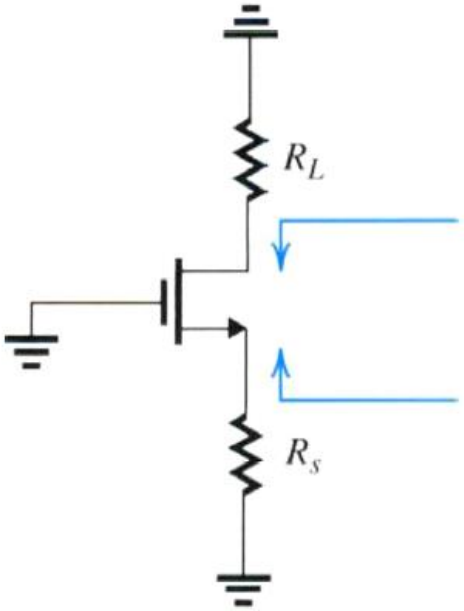
$$A_{vo} = -g_m r_o$$



$$A_v \equiv \frac{v_o}{v_i} = -g_{m1}(r_{o1} \parallel r_{o2})$$

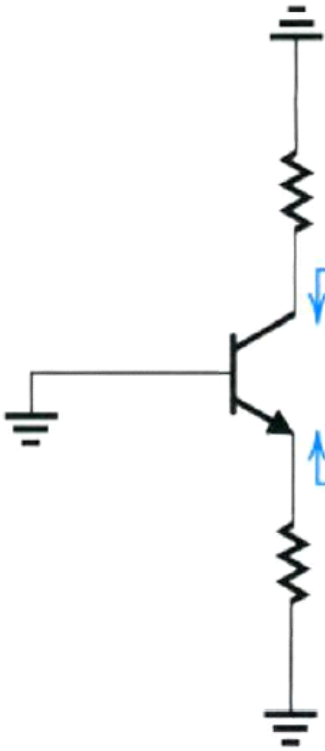
$$A_v = -\frac{1}{2} g_m r_o$$

Recap



$$R_{\text{out}} = r_o + R_s + g_m r_o R_s$$
$$\simeq r_o + (g_m r_o) R_s$$

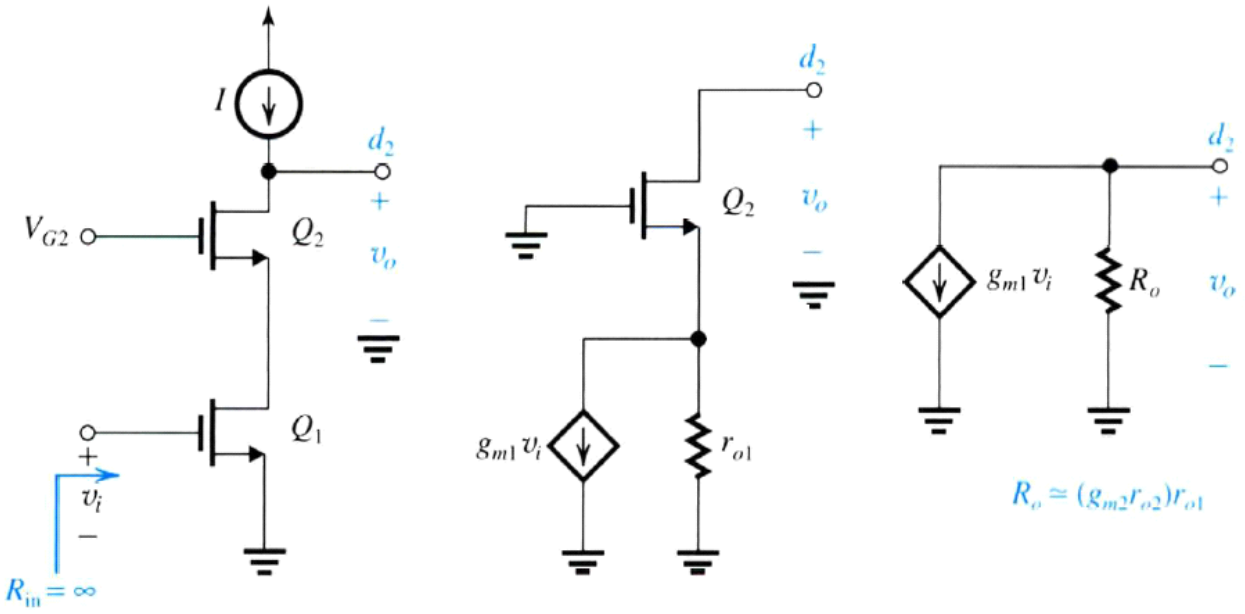
$$R_{\text{in}} = \frac{r_o + R_L}{g_m r_o + 1}$$
$$\simeq \frac{1}{g_m} + \frac{R_L}{g_m r_o}$$



$$R_{\text{out}} = r_o + (R_e \| r_\pi) + g_m r_o (R_e \| r_\pi)$$
$$\simeq r_o + (g_m r_o)(R_e \| r_\pi)$$

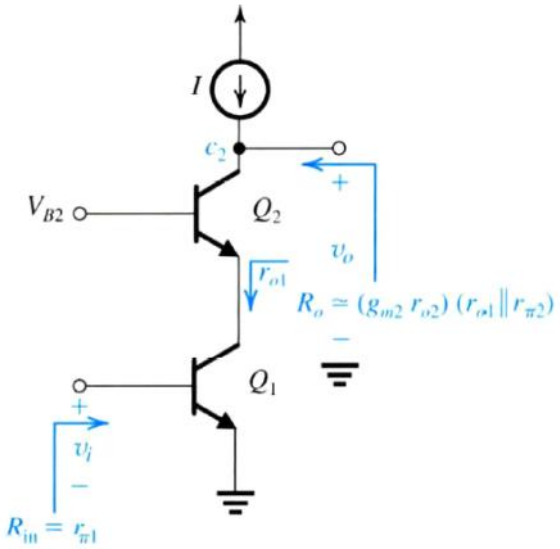
$$R_{\text{in}} = r_\pi \left\| \left[\frac{r_o + R_L}{g_m r_o + 1} \right] \right.$$
$$\simeq \frac{1}{g_m} + \frac{R_L}{g_m r_o}$$

Recap



$$A_{vo} \equiv \frac{v_o}{v_i} = -g_{m1} R_o$$
$$= -(g_{m1} r_{o1})(g_{m2} r_{o2})$$

$$A_{vo} = -(g_m r_o)^2 = -A_0^2$$

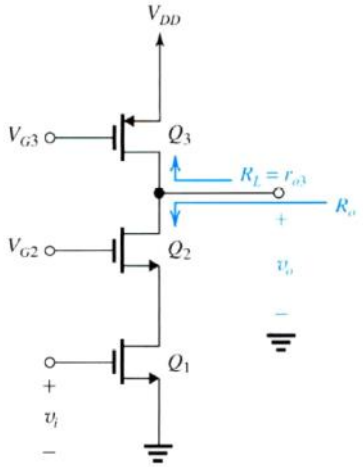


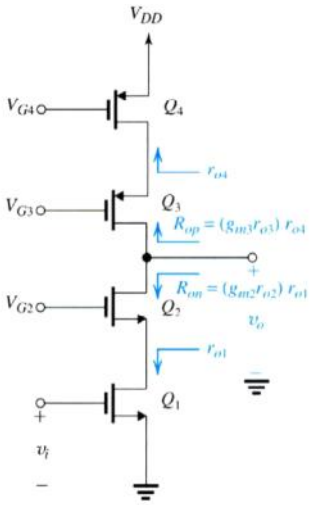
$$A_{vo} = -g_{m1}(g_{m2} r_{o2})(r_{o1} \| r_{\pi 2})$$

$$A_{vo} = -(g_m r_o)[g_m(r_o \| r_\pi)]$$

$$|A_{vo}|_{\max} = \beta g_m r_o = \beta A_0$$

Recap



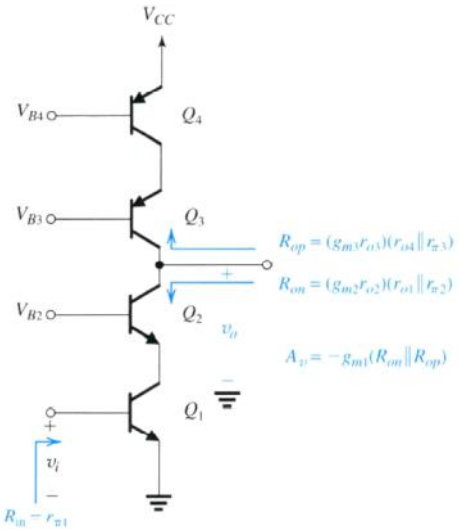


$$A_v = -g_{m1}(R_o \parallel R_L)$$
$$= -g_{m1}(g_{m2}r_{o2}r_{o1} \parallel r_{o3})$$

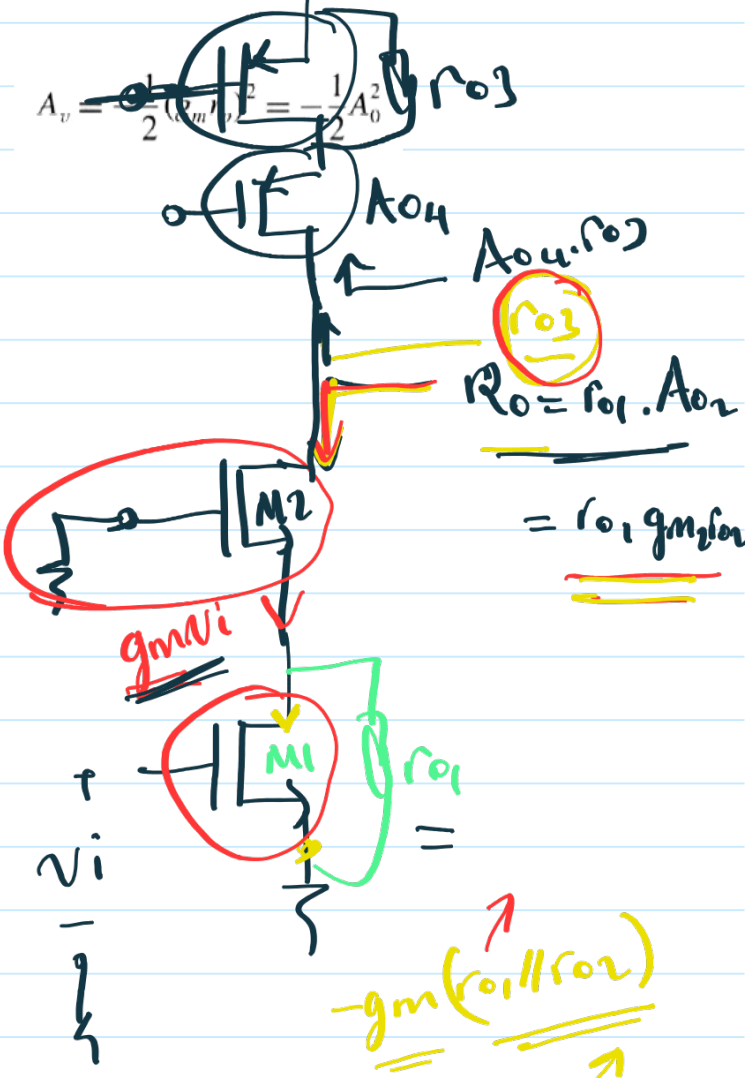
$$A_v \simeq -g_{m1}r_{o3}$$

$$A_v = -g_{m1}\{[(g_{m2}r_{o2})r_{o1}] \parallel [(g_{m3}r_{o3})r_{o4}]\}$$

$$A_v = -\frac{1}{2}(g_m r_o)^2 = -\frac{1}{2}A_0^2$$

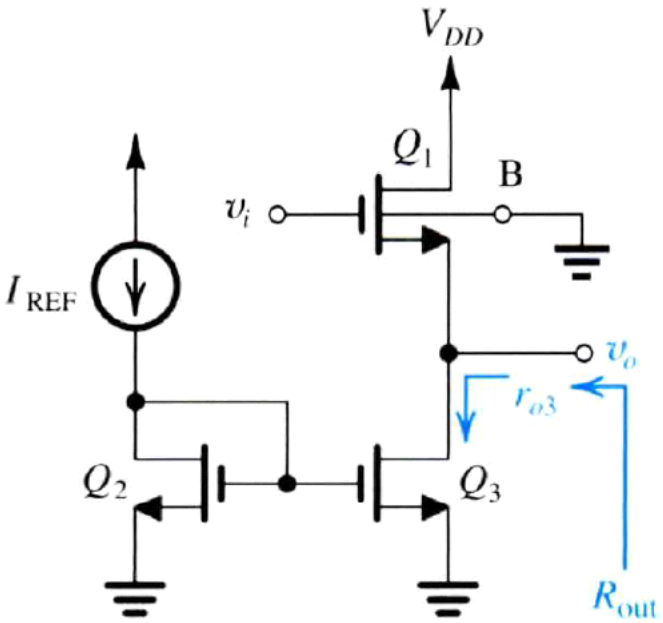


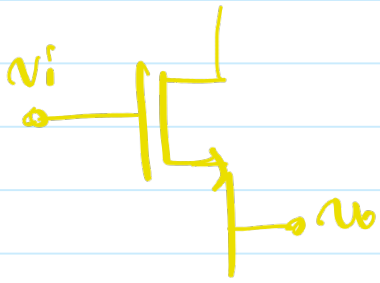
$-g_m r_{o3}$



$$-g_m \cdot (r_o g_m r_o \parallel r_o g_m r_o) = -\frac{A_o^2}{2}$$

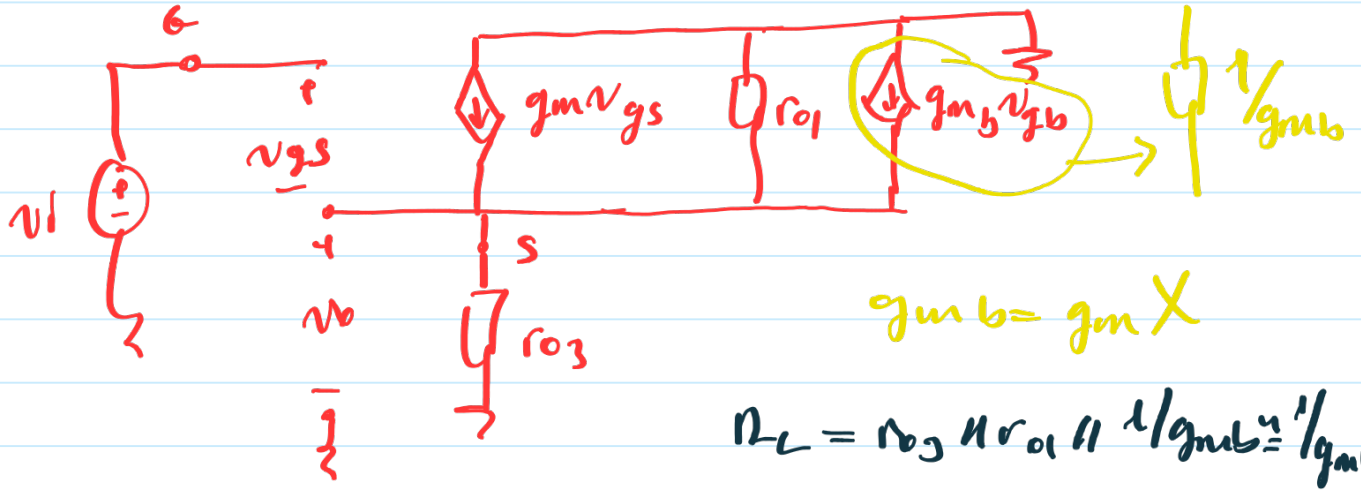
The IC Source Follower





$\frac{v_o}{v_i} \approx 1$

voltage buffer



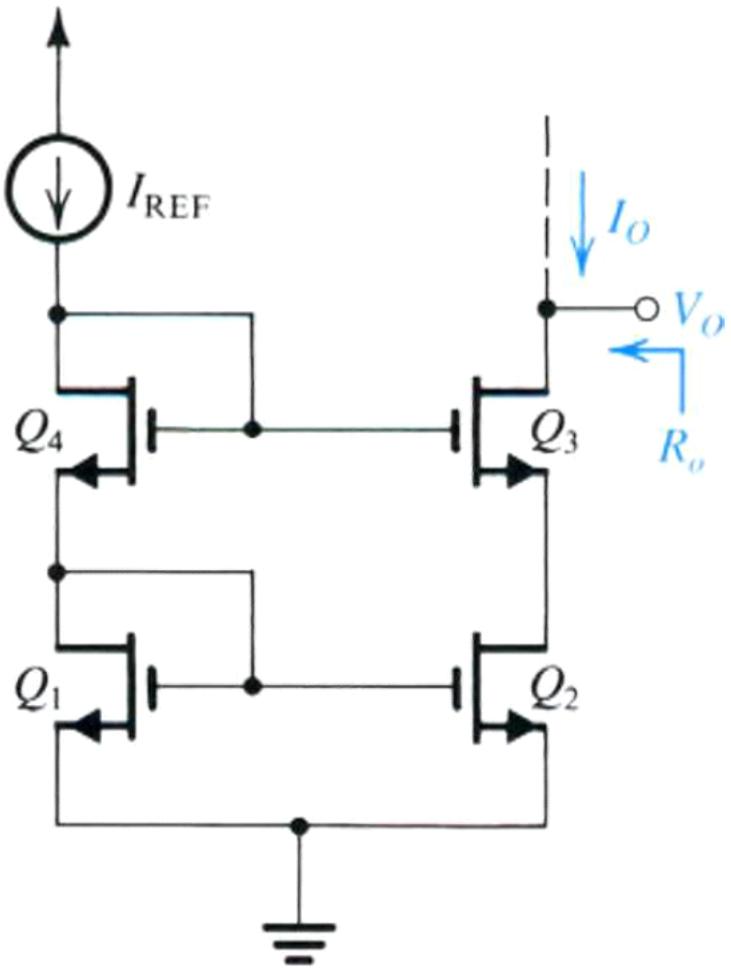
$g_{mb} = g_m \chi$

$R_L = r_{o3} \parallel r_{o1} \parallel 1/g_{mb} \cong 1/g_{mb}$

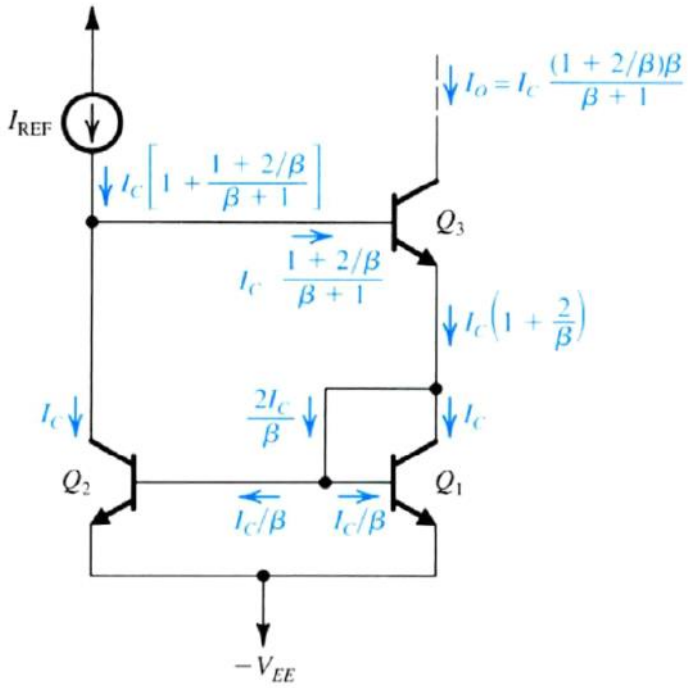
$$\frac{v_o}{v_i} = \frac{R_L}{R_L + 1/g_m} = \frac{1/g_{mb}}{1/g_{mb} + 1/g_m} = \frac{1}{1+\chi}$$

$$R_{out} = \frac{1}{g_m} \parallel r_{o1} \parallel r_{o3} \parallel \frac{1}{g_{mb}}$$

The Cascode Current Mirror

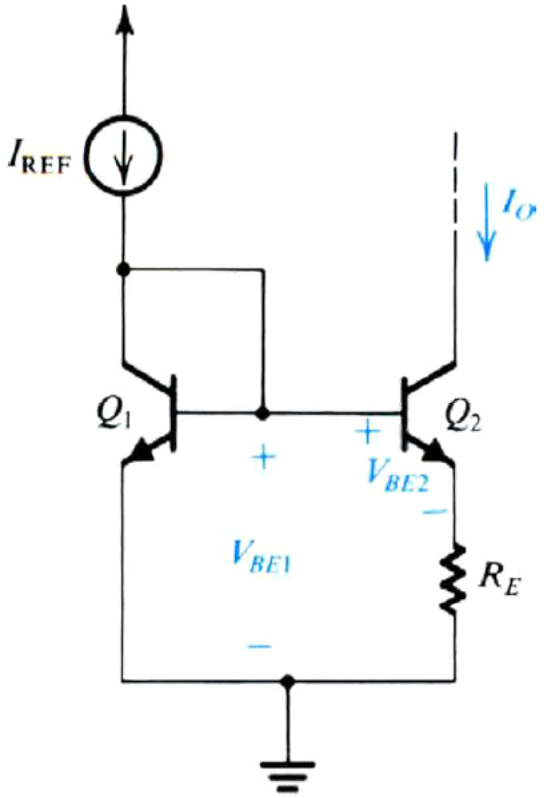


The Wilson BJT Current Mirror



The Wilson MOS Mirror

The Widlar Current Source



Template